The maximum-likelihood quantum amplitude estimation algorithm provides the best tradeoff between accuracy and circuit depth among quantum solutions for integral estimation

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***Abstract*—The approximation of an integral with numerical integration methods is a relevant problem in many fields, from finance to chemistry. Monte Carlo methods are a family of methods widely used for numerical integration but which are not always applicable to industrial tasks. Instead, there are quantum computing methods such as Quantum Amplitude Estimation able to offer a speed-up compared to the Monte Carlo methods for numerical integration which promise a significant impact, especially in finance. The Quantum Amplitude Estimation algorithm is a fundamental quantum algorithm with the potential to achieve a quadratic speed-up. Unfortunately, being competitive over classical Monte Carlo methods would require hardware performances currently not available. Therefore, alternative methods have been developed in order to exploit fewer resources while maintaining an advantageous theoretical scaling. We present a comparison between the standard QAE algorithm and two NISQ-friendly versions of the QAE on a numerical integration task. The Monte Carlo technique of Metropolis-Hastings instead is used as a classical benchmark. The scaling of the different algorithms are evaluated in terms of estimation error as a function of the number of samples, computational time, and length of the quantum circuits required by the quantum solutions. The conclusion highlights that an alternative approach is preferable with respect to employng the phase estimation routine. Indeed, the maximum likelihood estimation guarantees the best trade-off between the length of the quantum circuits and the precision in the integral estimation.**

***Index Terms*—Monte Carlo techniques, quantum amplitude estimation, integral estimation, NISQ devices**

I. Introduction

Numerical integration is a fundamental problem in many areas of science and engineering, and traditional numerical integration methods can be computationally expensive and require significant computational resources. Monte Carlo methods [1] have been widely used for numerical integration problems due to their flexibility and ability to handle high-dimensional problems. However, Monte Carlo methods suffer from the problem of statistical errors, which means that the accuracy of the approximation depends on the number of samples used in the simulation. Furthermore, such statistical methods suffer from the difficulty of generating a good sampling by starting from an arbitrary probability density function. As a result, Monte Carlo methods can be computationally expensive for high-precision integrations, making them less efficient for certain applications [2]. With the emergence of quantum computing [3], [4] the problem has been approached by using quantum algorithms, in particular after the introduction of the Quantum Amplitude Estimation (QAE) algorithm [5]. QAE is a quantum algorithm that can be used to estimate the amplitude of a specific state in a quantum system with exponential speed-up compared to classical algorithms. This allows QAE to be used for efficient numerical integration of complex functions. In particular, QAE has been shown to be able to beat the classical Monte Carlo methods for numerical integration, especially in high-dimensional integration problems. By using QAE, it is

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possible to estimate the integrals with high precision using fewer samples, thus significantly reducing the computational resources required for numerical integration. Moreover, QAE can be used to solve a wide range of numerical integration problems, including those in finance [6], physics [7], [8], and machine learning [9]–[12]. It has been shown that QAE can be applied to estimate option prices [13] in financial derivatives [14], [15], solve differential equations [16], and perform quantum simulation, among others. The QAE algorithm promises quadratic speedup compared to classic Monte Carlo techniques. Such advantage, however, is constrained by the existence of a fault tolerant quantum hardware [17], [18] that does not exist today. In currently available noisy intermediate scale quantum (NISQ) devices [19], the QAE algorithm has not yet shown an advantage in practical cases [20] due to the low coherence times of the qubits and because the error carried by the gates. Indeed the QAE circuit consists in 2^M applications of controlled quantum queries and a quantum Fourier transformation that are not suitable for an implementation on such device. Therefore, alternative approaches have been proposed for QAE that involve a lower-depth circuit. Such quantum-classical hybrid algorithms can be divided by two categories, namely the Maximum Likelihood Amplitude Estimation (MLAE) [21] approach and the Iterative QAE [22] approach, respectively. Here, the standard QAE algorithm, the MLAE, the IQAE and the classical Metropolis-Hastings Monte Carlo (MHMC) algorithm are compared. The scaling on the estimation error, the execution time and the depth of the circuits are considered. An 1D integral was chosen as a benchmark, simple in terms of the depth of the required quantum circuit but representative of the problem. All simulations were performed on the quantum circuit simulator *aer_simulation* of Qiskit library, by involving 3 qubits. The Iterative hybrid approach has a depth of the circuits that implement it on average higher and also in terms of scaling with respect to the estimation error it requires ever deeper circuits than the MLAE approach. Instead, the results show that the two approaches share a comparable speed-up. The paper is therefore divided into sections as follows: Section II defines the QAE and its application to the numerical integration problem, Section III presents the NISQ-friendly alternatives of the QAE and Section IV presents the results of the simulations.

II. Original quantum amplitude estimation algorithm

In this Section, the original version of the QAE, for numerical integration, is summarized. In order to introduce it, let's first consider an integral

$$I = \int_X f(x)\, dx \tag{1}$$

and let's define a probability distribution $p(x)$ such that $p(x) \neq 0$ for $x \in X$ than the integral can be rewritten as

$$I = \int p(x)g(x)\, dx = \mathbb{E}_p[g(x)] \tag{2}$$

where $g(x) \equiv f(x)/p(x)$ in X and 0 otherwise. Classicaly, the Monte Carlo method consists of a set of methods to estimate the integral in the Equation 2. Such methods start from the assumption of sampling M values $(x_0, \ldots, x_{M-1})$ from the $p(x)$ and then estimate I with the average estimator.

The estimation error bound of classical MC simulation scales as $\mathcal{O}(1/\sqrt{M})$, where M denotes the number of (classical) samples. Instead, the quantum algorithm called QAE allows to achieve a quadratic speed-up as explained below. The QAE problem can be stated as follows: given an operator $\mathcal{A}$ acting on $n+1$ qubits such that

$$\mathcal{A}\left|0\right\rangle_n \left|0\right\rangle = \sqrt{1-a}\left|\Psi_0\right\rangle\left|0\right\rangle + \sqrt{a}\left|\Psi_1\right\rangle\left|1\right\rangle \tag{3}$$

where $a \in [0,1]$ is the probability to measure the ancilla qubit in 1 and it is unknown [5]; $\left|\Psi_0\right\rangle$ and $\left|\Psi_1\right\rangle$ are two normalized states, not necessarily orthogonal [22].

To relate the unknown value a to the integral in the Equation 2 one can define

$$\left|\Psi_1\right\rangle = \frac{1}{\sqrt{a}} \sum_x \sqrt{p(x)g(x)}\left|x\right\rangle_n \tag{4}$$

therefore $a \sim \mathbb{E}_p[g(x)]$.

Let's define an operator $\mathcal{Q} = \mathcal{A}\mathcal{S}_0\mathcal{A}^\dagger\mathcal{S}_{\Psi_0}$ where $\mathcal{S}_0 = \mathbb{I} - 2\left|0\right\rangle_{n+1}\left\langle 0\right|_{n+1}$ representing reflection with respect to the state $\left|0\right\rangle$ while $\mathcal{S}_{\Psi_0} = \mathbb{I} - 2\left|\Psi_0\right\rangle_n\left\langle\Psi_0\right|_n \otimes \left|0\right\rangle\left\langle 0\right|$ the reflection with respect to the state $\left|\Psi_0\right\rangle_n$. Applications of $\mathcal{Q}$ are denoted as *quantum samples* or *oracle queries* [22].

The canonical QAE follows the form of quantum phase estimation (QPE): (i) it uses m ancilla qubits, initialized in equal superposition, to represent the final result, (ii) it defines the number of quantum samples as $M = 2^m$ and (iii) it applies geometrically increasing powers of $\mathcal{Q}$ controlled by the ancillas. Eventually, it performs an inverse quantum Fourier transform (QFT) on the ancilla qubits before they are measured. The measurement of m ancilla qubits produces a bit string equivalent to an integer $y \in \{0, \ldots, M-1\}$ and an estimation of a can be defined as $\tilde{a} = \sin^2\theta_a$ where $\theta_a \equiv y\pi/M$. The algorithm produces an estimation $\tilde{a}$ such that

$$|a - \tilde{a}| \leq \delta \equiv \frac{2\pi\sqrt{a(a-1)}}{M} + \frac{\pi^2}{M^2} \sim \mathcal{O}\left(\frac{1}{M}\right) \tag{5}$$

with a probability

$$\begin{aligned}\mathbb{P}[y||a-\tilde{a}| \leq \delta] &= \mathbb{P}[y = \lfloor M\theta_a/\pi \rfloor] + \mathbb{P}[y = \lceil M\theta_a/\pi \rceil] \\ &= \frac{\sin^2(M\Delta)}{M^2\sin(\Delta)} + \frac{\sin^2(\pi - M\Delta)}{M^2\sin(\pi/M - \Delta)} \geq \frac{8}{\pi^2}\end{aligned} \tag{6}$$

where $M \leq 2$ while Δ is the minimal distance on the unit circle between the angles θ_a and $\pi y/M$. Therefore returning a scaling complexity of $\mathcal{O}(1/M)$ for M applications of quantum samples $\mathcal{Q}$.

The success probability to obtain $\tilde{a}$ with the discrepancy shown in the Equation 5 can quickly be boosted to close to 100% by repeating the QAE circuit multiple times and by using the median estimate. Since the success probability of QAE

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is larger than $8/\pi^2$ we would in principle only need, for instance, 24 repetitions to achieve a success probability of 99.75% [20]. However, current quantum hardware introduces additional errors. In the work [20] the circuit was repeated 8192 times to get a reliable estimate of a.

III. Alternative Quantum Amplitude Estimation Methods

As discussed, today's hardware limitations prevent one from getting the promised speedup. For this reason, various algorithms have been developed involving both classical and quantum devices in order to reduce the resources required for quantum computation without renouncing a speedup against Monte Carlo methods. The alternative hybrid algorithms can be divided into two categories. The first category (I), to which we will refer with a *Maximum Likelihood Amplitude Estimation (MLAE) approach*, consists of a Grover-like circuit, and a maximum likelihood estimation on a classical processor. Such approach has been discussed by different groups: in Ref. [21] the idea was introduced for the first time, while two variants have been proposed later [23], [24].

The second category (II), under the name of *iterative approach*, consists of an alternating sequence of Grover-like circuits, where the oracle query $\mathcal{Q}$ is applied k times, and a classical algorithm to compute the value k of $\mathcal{Q}$ applications for the next quantum iteration. Such iterative procedure leads to decrease the confidence interval of the estimation $\tilde{a}$ up to the error ϵ with up to $T = \lceil \log_2(\pi/8\epsilon) \rceil$ iterations [22]. Such algorithm in turn takes inspiration from the variant of the quantum approximation counting described in Ref. [25].

A. MLAE approach

As mentioned earlier, such a quantum-classical hybrid class of algorithms involves both a Grover-like circuit and a Maximum Likelihood Estimation (MLE) [26] on a classical processor, respectively. In the following, the strategy of the algorithm is analyzed. Let $\mathcal{A}$ be an operator such that

$$\mathcal{A} \left|0\right\rangle_n \left|0\right\rangle = \cos\theta_a \left|\Psi_0\right\rangle \left|0\right\rangle + \sin\theta_a \left|\Psi_1\right\rangle \left|1\right\rangle \tag{7}$$

The aim of the algorithm is to estimate $a = \sin^2\theta_a$. It is possible to replace QPE with a set of Grover iterations combined with a Maximum Likelihood Estimation (MLE) [21].

Let's perform N_k sampling from the circuit

$$\begin{aligned} &\mathcal{Q}^k \mathcal{A} \left|0\right\rangle_n \left|0\right\rangle = \\ &= \cos\left((2k+1)\theta_a\right) \left|\Psi_0\right\rangle \left|0\right\rangle + \sin\left((2k+1)\theta_a\right) \left|\Psi_1\right\rangle \left|1\right\rangle \end{aligned} \tag{8}$$

From the Eq. 8 we are able to model the probability distribution of the measurements of the ancilla qubit with $\mathbb{P}[|0\rangle] = \cos^2\left((2k+1)\theta_a\right)$ and $\mathbb{P}[|1\rangle] = \sin^2\left((2k+1)\theta_a\right)$.

The likelihood of such model is

$$L_k(h_k, \theta_a) = \left[\sin^2\left((2k+1)\theta_a\right)\right]^{h_k} \left[\cos^2\left((2k+1)\theta_a\right)\right]^{N_k - h_k} \tag{9}$$

where h_k is the number of the is the number of measurements that have returned 1 and N_k is the total number of measurements.

Next, let's perform the sampling with $M+1$ times with $k = 0, \ldots, M$ in order to compute the total likelihood

$$L(h, \theta_a) = \prod_{k=0}^{M} L_k(h_k, \theta_a) \tag{10}$$

Using the maximum likelihood procedure we are able to compute the optimal θ_a such that our model approximates the true distribution as closely as possible

$$\hat{\theta}_a := \arg\max_{\theta_a} L(h, \theta_a) = \arg\max_{\theta_a} \ln L(h, \theta_a) \tag{11}$$

Now a and θ_a are uniquely related through $a = \sin^2\theta_a$ in the range $0 \leq \theta_a \leq \pi/2$ so $\hat{a} = \sin^2\hat{\theta}_a$, as the function is invertible in such interval. As shown in Ref. [24] a pure MLAE method as just described above brings $less$ than a quadratic advantage respect to the classical Monte Carlo. Always in [24] an algorithm is described that exploits the MLAE method but alternated with the variational optimization step.

B. Iterative approach

As in the previously discussed approach, the IQAE replaces the standard QAE circuit, with the low complexity circuit defined in Equation 8. The difference between the two approaches is that, while in the MLAE method the k times with which $\mathcal{Q}$ is applied ranges from 0 to M, in the iterative approach k is computed by a classical routine which takes as input a confidence interval $[\theta_l, \theta_u]$ for the angle θ_a.

The algorithms described in Refs. [22], [25], [27] belong to the iterative approach class. In the following, the most representative IQAE is described in details [22]. The IQAE uses the quantum computer to approximate, with N_{shots} measure, $\mathbb{P}[|1\rangle] = \sin^2\left((2k+1)\theta_a\right)$ for the last qubit in $\mathcal{Q}^k \mathcal{A} \left|0\right\rangle_n \left|0\right\rangle$ for different powers k. The first step consists to set a confidence interval $[\theta_l, \theta_u] \subseteq [0, \pi/2]$. For convenience we set $[\theta_l, \theta_u] = [0, \pi/2]$.

After that we must define a confidence level $1 - \alpha \in (0, 1)$, a target accuracy $\epsilon > 0$ and a number of shots $N_{shots} \in \{1, \ldots, N_{max}(\epsilon, \alpha)\}$ where

$$N_{max}(\epsilon, \alpha) = \frac{32}{(1 - 2\sin(\pi/14))^2} \log\left(\frac{2}{\alpha} \log_2\left(\frac{\pi}{4\epsilon}\right)\right) \tag{12}$$

From the definition of the inputs $(\epsilon, \alpha, N_{shots})$ one can calculate the maximum possible error

$$L_{max}(\epsilon, \alpha, N_{shots}) = \arcsin\left(\frac{2}{N_{shots}} \log\left(\frac{2T(\epsilon)}{\alpha}\right)\right)^{1/4} \tag{13}$$

For each iteration i an integer k_i must be defined. In Ref. [22] it is defined a routine called FindNextK which takes the integer k_{i-1} ($k_0 = 0$) and the interval $[\theta_l, \theta_u]$ obtain from the previously iteration. In such core routine of the algorithm, the idea is to define an integer k_i such that $[(4k_i+2)\theta_l, (4k_i+2)\theta_u]_{mod 2\pi}$ is fully contained either in $[0, \pi]$ or $[\pi, 2\pi]$. That choice follows the aim of estimating, instead of $\sin^2\left((2k+1)\theta_a\right)$, the quantity $\cos\left((4k+2)\theta_a\right)$ which can be inverted only in $[0, \pi]$ or $[\pi, 2\pi]$.

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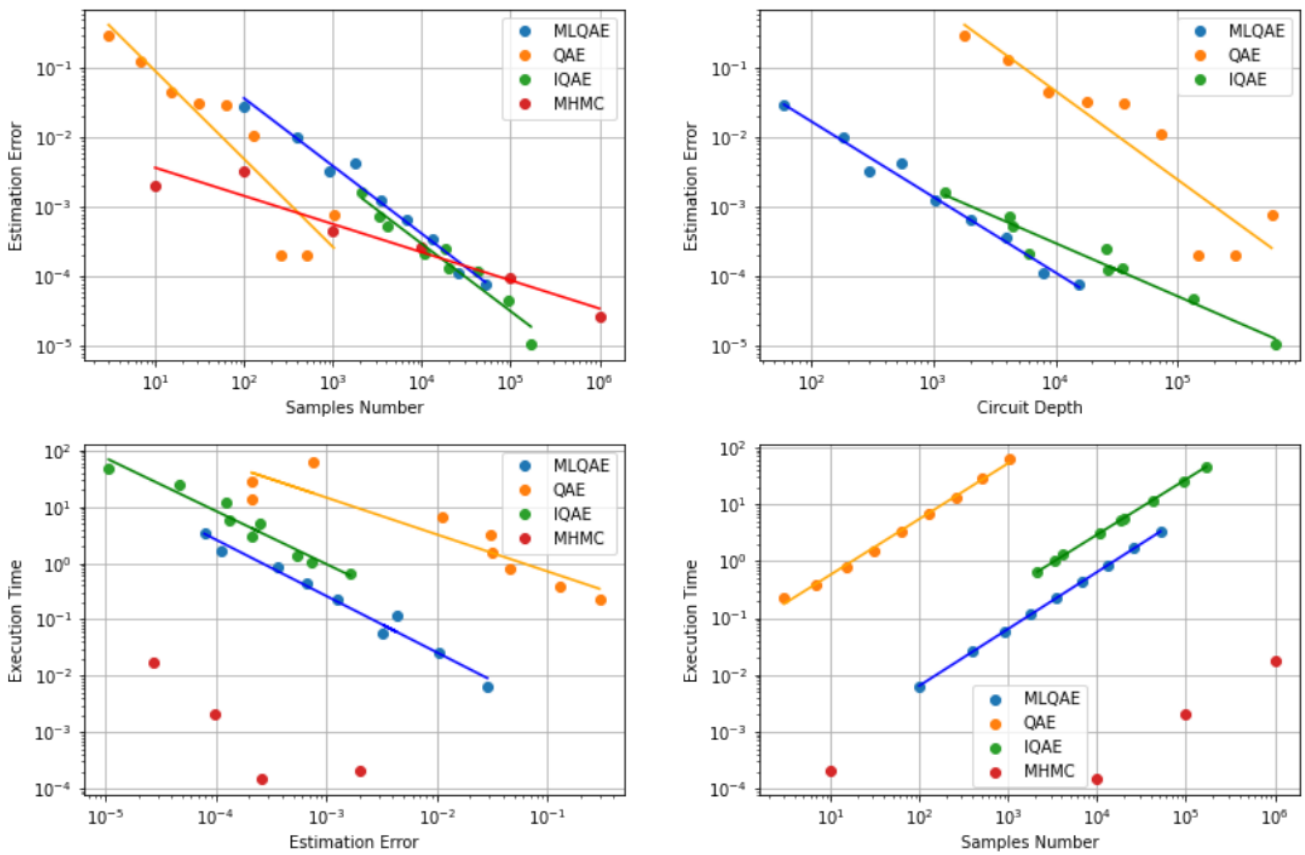

Fig. 1. Performance of the amplitude estimation algorithms to estimate an 1D integral with $n = 3$ qubits. All results of the quantum algorithms were obtained with the local qiskit simulator *aer_simulation* and each plot shows the average of 10 executions with different simulator seeds. All simulations of the IQAE [22] algorithm were performed with a 90% confidence interval. The top-left plot shows how much the error on estimating the integral decreases as a function of the number of samples/oracle queries. The data are fitted with a function $x^{-\alpha}$ in loglog scale. For the algorithms MLQAE [21], standard QAE, IQAE, and classical Metropolis-Hastings Monte Carlo (MHMC) the slopes are respectively of -0.974 ± 0.058, -1.267 ± 0.206, -0.971 ± 0.092 and -0.485 ± 0.051 (result obtained without considering the case with 10 samples). MLQAE and IQAE are performed with 100 shots per quantum circuit while the QAE was executed standalone. The data show quadratic speedup for the QAE (orange) and slightly less than quadratic speedup for the MLQAE and IQAE algorithms (blue and green, respectively) compared to estimation as a classic sampling Monte Carlo method (red). The top-right plot shows the estimation error as a function of the higher circuit depth (we must consider the highest one because MLQAE and IQAE required the simulation of more than one quantum circuit). The two lower plots show instead the execution time (for the quantum algorithms it is the execution time of the QisKit local simulator) in function of the estimation error (left) and the number of samples/oracle queries (right).

If FindNextK finds an integer k_i which satisfies the constraints, then one performs the N shots (which depends of N_{shots}) of the circuit $\mathcal{Q}^k \mathcal{A} |0\rangle_n |0\rangle$ to estimate $a_i = \mathbb{P}[|1\rangle]$. Otherwise the iteration i must be performed with $k_i = k_{i-1}$.
The next step consists to define an interval $[a_i^{min}, a_i^{max}]$ using different techniques such as the Chernoff-Hoeffding method according to which $a_i^{max} = \min(1, a_i + \epsilon_{a_i})$ and $a_i^{min} = \max(0, a_i - \epsilon_{a_i})$ where

$$\epsilon_{a_i} = \sqrt{\frac{1}{2N} \log\left(\frac{2T}{\alpha}\right)} \tag{14}$$

The new confidence interval $[\theta_l, \theta_u]$ is than calculated from $[a_i^{min}, a_i^{max}]$ reversing the formula $a = \cos((4k+2)\theta_a)$. Such algorithm demonstrates a quadratic speedup with respect to Monte Carlo techniques up to a factor given by N_{max}. The speedup in such case depends by the confidence level which can be arbitrarily set.

IV. Bench-marking the methods of quantum amplitude estimation

In order to benchmark the effectiveness of algorithms of quantum amplitude estimation, we identify an integral that is particularly representative thanks to small depth required for the implementation on a gate-model quantum computer. More specifically selected function consists of $\sin^2 x$. We therefore consider the following integral

$$\mathcal{I} = \frac{1}{b_{max}} \int_0^{b_{max}} \sin^2 x \ \ dx \tag{15}$$

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QAE methods list							
Algorithm	NISQ-readiness	Qubits	Depth	N_{shots} vs α	Speedup over MC	Ref.	Type
QAE	Low	$n+1+\log 1/\epsilon$	$d \cdot 1/\epsilon + \log\log 1/\epsilon$	$B(p=0,81, k=1, n=N_{shots})$ **	$1/\epsilon$	[5]	O
QAE NO-PE *	Medium	$n+1$	$d \cdot 1/\epsilon$	–	$\epsilon^{-4/3}$	[21], [28]	I
MLQAE *	High	$n+1$	$< d \cdot 1/\epsilon$	–	$> \epsilon^{-4/3}$	[24]	I
Power-law AE *	High	$n+1$	$d \cdot (1/\epsilon)^{1-\beta}$	–	–	[23]	I
IQAE	Medium	$n+1$	$d \cdot 1/\epsilon$	N_{max} Eq. 12	$N_{max} \cdot 1/\epsilon$	[22]	II
SQAE	Medium	$n+1$	–	–	–	[25]	II
FAE	Medium	$n+1$	$d \cdot 1/\epsilon$	–	$(1/\epsilon) \cdot \ln log(\pi/\epsilon)$	[27]	II

* Classical optimization required
** Binomial distribution

TABLE I
COMPARISON BETWEEN QAE ALGORITHMS. HERE $n+1$ IS THE NUMBER OF QUBITS ON WHICH THE ORACLE QUERY $\mathcal{Q}$ IS APPLIED, d IS THE DEPTH OF $\mathcal{Q}$ WHILE ϵ IS THE TARGET ACCURACY AND α THE CONFIDENCE LEVEL. IN THE LAST TWO ALGORITHMS $\beta \in (0,1]$, $k \leq 2$ AND $q \in [1,\ldots,k-1]$. IN THE LAST COLUMN IT IS INDICATED TO WHICH APPROACH THE ALGORITHM BELONGS TO, BASED ON THE CLASSIFICATION OF SECTION III: O CORRESPONDS TO THE ORIGINAL QAE, I TO THE MLAE APPROACH AND II TO THE ITERATIVE RESPECTIVELY

which is approximated by the following summation of 2^n samples

$$\mathcal{D} = \sum_{x=0}^{2^n-1} \frac{1}{2^n} \sin^2\left(\frac{(x+1/2)\, b\text{max}}{2^n}\right) \xrightarrow{n\to\infty} \mathcal{I}. \tag{16}$$

encoded by a qubit register q of n qubits.
The integral $\mathcal{D}$ can be interpreted as the expectation value of the function

$$f(x) = \sin^2\left(\frac{(x+1/2)\, b\text{max}}{2^n}\right) \tag{17}$$

with an uniform probability distribution $p(x) = 1/2^n$.
The corresponding quantum circuits $\mathcal{P}$ to encode $p(x)$ on the $n+1$ qubit register (consisting in q plus an ancilla qubit) and $\mathcal{R}$ to encode $f(x)$ are

$$\mathcal{P}\left|0\right\rangle_n \left|0\right\rangle = \frac{1}{\sqrt{2^n}} \sum_x \left|x\right\rangle_n \left|0\right\rangle \tag{18}$$

$$\mathcal{R}\left|x\right\rangle_n \left|0\right\rangle = \left|x\right\rangle_n \left(\sin\left(\frac{(x+1/2)\, b\text{max}}{2^n}\right)\left|1\right\rangle + \cos\left(\frac{(x+1/2)\, b\text{max}}{2^n}\right)\left|0\right\rangle\right) \tag{19}$$

respectively [21]. $\mathcal{P}$ can be realized with Hadamard gates, and $\mathcal{R}$ with controlled-Y rotation gates. The methods described in the previous Section aim to reduce the number of qubits and the length of the circuits while maintaining an advantage over the classic Monte Carlo method.
In the Table I the QAE variants are classified on the basis of the relative NISQ readiness. Since the number of the qubits and the circuits depth are the only two parameters to evaluate the NISQ readiness, we can only make a relative classification by starting from classifying the standard QAE algorithm by conventionally assigning a low NISQ readiness. However, the algorithm scaling does not depend only on the circuits depth but also on the number of shots N_{shots} and potentially on the number of calls to the QPU. Another parameter to consider is the confidence level α, i.e. the probability of obtaining a result with the desired target error ϵ. In the original QAE the lowest α is 81%, but it can be increased by performing more parallel circuit runs, which means a higher N_{shots}. Other algorithms, like the IQAE, allow one to arbitrarily choose the confidence level therefore there is a correlation between the computational cost and α. All the algorithms described above succeed in estimating an integral with an error that scales as $\mathcal{O}(x^{-\alpha})$ where x is the number of samples, where α has been empirically evaluated, as follows.

V. DISCUSSION

From the fit on the data in Figure 1 we obtain the theoretically predicted trends with an estimate that is always within 1.3σ. For the standard QAE algorithm (theoretical $\alpha = 1$), we estimate $\alpha = -1.267 \pm 0.206$ while for the IQAE and MLAE algorithms we estimate a value compatible with $\alpha = 1$ and thus with the QAE algorithm (a trend with $\alpha < 1$ or at least slightly less performing than the standard QAE is expected). One should consider at the same time another figure of merit is represented by circuit depth, since it impacts the viability on modern quantum devices in the NISQ era.
IQAE and MLAE are classical-quantum hybrids that are therefore compositions of runs of multiple circuits interspersed with classical routines. The Figure 1 shows the maximum depths reached by the circuits to perform an estimation of the integral. We note how the two hybrid algorithms IQAE and MLAE - as expected - have a shallower depth than the standard QAE but in particular the IQAE algorithm needs very long circuits to perform an estimation of the same quality as MLAE.

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In fact, even though the number of samples is similar between the two algorithms at the same error ϵ , because of the structure of IQAE, interactions with multiple oracle calls (which are counted as samples) concatenated in the same circuit can happen. Such property is also reflected in the execution time of the algorithms by quantum simulator. Indeed, it turns out that the MLQAE algorithm is the fastest at the same error of integral estimation ϵ. Such conclusion is general for any number of qubits. The iterative method, as well as the MLQAE, performs on the quantum processor the estimate of the probability $\mathbb{P}[|1\rangle]$ on the ancilla qubit which will depend on the unknown value of the integral and not on the number of qubits dedicated to encoding the integral. The two lower plots in Figure 1 allow us to extrapolate the execution times on the simulator of these quantum solutions. We see that generating a sample with the Monte Carlo method takes 4 orders of magnitude less time than MLQAE. Around the interval $\left[10^{-13}, 10^{-11}\right]$ we could find an advantage in terms of computational time of the MLQAE algorithm. Approximately 10^{24} classical samples are required to achieve precision in such range. Such values are lower if we consider a physical implementation of a quantum computer. If we assume execution of the quantum gates 100 times faster than its simulated counterpart, then the advantage can be obtained around the interval $\left[10^{-9}, 10^{-8}\right]$ with a number of classical samples required around 10^{-17}. In the case of gate execution 1000 times faster than the laptop simulation then this advantage could be in the range $\left[10^{-7}, 10^{-6}\right]$ where it would take about 10^{11} classical samples. However, these evaluations are not complete since they do not take into account three factors: the noise to which the NISQ devices are subject, the number of qubits which affects the discretization of the integral and which leads to a systematic error of the estimation and, finally, the dilemma related to algorithms with a quadratic speedup which, if implemented with an error correction method, can lose. Further analysis of the resources required by QAE algorithms is reported in [8], based on a practical example of a multidimensional integral. To achieve a precision of the order of 10^{-3}, 1000 qubits are estimated to be used, combined with long consistency times and high fidelity gate.

VI. Conclusions

The scaling of the different versions of the QAE algorithms have been evaluated in terms of how the estimation error decreases as a function of the number of samples, computational time, and length of the quantum circuits. The maximum likelihood QAE, which does not make use of phase estimation, results as the best candidate to be first to demonstrate an advantage over the classical Monte Carlo methods. The conclusion is supported by the trade-off between the length of the quantum circuits and the precision in the integral estimation.

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